

PM Movers Pro

A professional Pine Script v6 premarket gap-and-volume screener, with implementation
whitepaper and tutorial

Deliverable	Improved Pine Script screener plus operating manual
Primary use	U.S. equity premarket watchlist scanning, 04:00-09:30 ET by default
Version	PM Movers Pro v2 - Pine Script v6
Date	April 24, 2026

This document is educational and technical. It does not recommend any security, trade, or portfolio action. Extended-hours markets may have lower liquidity, wider spreads, and higher volatility than regular trading hours.

1. Executive Summary

Your original screener correctly targeted the core idea: detect premarket movers using a gap threshold and relative volume threshold. The upgraded version keeps that workflow, but hardens the implementation for Pine Screener use, adds direction-aware logic, improves session hygiene, and exposes more useful scan columns.

The most important design change is the default volume baseline. Pine Screener is excellent for watchlist scanning, but it evaluates a limited dataset and applies request-call/timeframe restrictions. Because of that, a multi-day intraday premarket-volume array can be unreliable in the screener, especially on lower timeframes. PM Movers Pro defaults to screener-safe ADV-based volume modes and keeps the PM-to-PM RVOL mode for chart use when sufficient extended-hours history is available.

Key outcomes:

- Migrated to Pine Script v6 and reorganized inputs using professional naming and grouping conventions.
- Added screener-safe volume filters: PM volume / ADV and PM pace / ADV.
- Fixed down-mover logic by tracking PM low, not only PM high and PM last.
- Added daily reset logic to avoid stale held values before the next premarket session.
- Added quality filters: minimum price, PM volume, average daily volume, and PM dollar volume.
- Prioritized the first ten plots for Pine Screener columns and filters.
- Added separate up/down first-trigger alerts, chart markers, and a last-bar dashboard.

2. Research Basis and Platform Constraints

The implementation was updated against current TradingView and market-structure documentation. The following constraints guided the engineering choices:

Constraint	Research finding	Design response
Pine Screener outputs	Pine Screener scans watchlists using indicator plots and alert conditions. Only plots from the selected indicator can be table columns; the first ten plots are added initially.	The script places the most important outputs first: PM Mover, Direction, PM %, Volume Metric, PM Vol/ADV, PM RVOL, PM volume, PM dollar volume, Score, Config OK.
Pine Screener request limits	Pine Screener supports request.*() functions, but limits scripts to no more than five separate request.*() calls and only supports a fixed set of request timeframes.	The default mode uses two daily request.security() calls only: prior daily close and completed average daily volume.
Screener dataset length	Pine Screener performs calculations on the last 500 bars.	The chart-built PM-to-PM average remains available, but it is not the default screener mode.

Session/timezone handling	Pine time-based sessions use HHmm-HHmm strings and can return na outside a session; IANA timezone strings automatically handle daylight saving time.	Session detection uses time(timeframe.period, session, timezone) and defaults to America/New_York.
U.S. premarket window	Nasdaq defines Pre-Market Hours as 4:00 a.m. to 9:30 a.m. on business days; NYSE Arca Equities also shows an early trading session from 4:00 a.m. to 9:30 a.m. ET.	Default session is 0400-0930 ET, with an editable input because broker and venue access can differ.

3. Baseline Assessment of the Original Screener

The original Pine v5 script had a solid foundation: it detected a premarket session, requested the prior daily close, accumulated premarket volume, tracked PM high and last price, calculated a gap percentage, stored prior PM totals in an array, plotted screener columns, and exposed alert conditions.

The main limitations were practical rather than conceptual:

- The PM-to-PM average depends on having enough historical premarket bars available in the script runtime. This is fragile in Pine Screener because the screener uses only the last 500 bars.
- Down movers were not direction-aware when using the high-so-far price basis. A true short-side premarket mover should evaluate PM low or last price.
- Held values could remain visible across the midnight-to-04:00 interval unless the next PM session had already reset them.
- The volume threshold had one interpretation even though daily-volume baselines and PM-to-PM baselines represent different quantities.
- There were no liquidity guards for minimum price, minimum PM volume, minimum ADV, or PM dollar volume.
- The script had only one generic marker style and did not separate up/down alert conditions.

4. System Architecture

PM Movers Pro is designed as a deterministic screening pipeline. It avoids broad symbol requests and keeps all screening logic local to the current chart symbol.

Stage	Inputs / Calculations	Outputs
1. Session state	Premarket session string, timezone, weekday filter, day reset	inPm, pmStart, pmEnd, elapsed minutes
2. Reference data	Prior daily close and completed daily average volume via 1D request.security()	prevClose, avgDailyVol
3. PM accumulation	Cumulative PM volume, high, low, last, dollar volume	pmCumVol, pmHigh, pmLow, pmLast, pmDollarVolume

4. Metric engine	Gap %, PM Vol/ADV, PM Pace/ADV, chart PM RVOL	selectedPct, volumeMetric, signalScore
5. Filters	Gap threshold, volume mode threshold, price/volume/dollar-volume quality gates	moverUpNow, moverDownNow
6. Publishing	Plots, alertcondition(), markers, dashboard	Pine Screener columns, alerts, chart feedback

Core formulas

Metric	Formula	Interpretation
PM High %	$(\text{PM high} - \text{prior close}) / \text{prior close} * 100$	Best upside extension from prior close during premarket.
PM Low %	$(\text{PM low} - \text{prior close}) / \text{prior close} * 100$	Best downside extension from prior close during premarket.
PM Last %	$(\text{PM last} - \text{prior close}) / \text{prior close} * 100$	Current/last PM change versus prior close.
PM Vol / ADV	PM cumulative volume / completed average daily volume	Screener-safe participation metric; default threshold is 0.10.
PM Pace / ADV	$(\text{PM cumulative volume} / \text{elapsed session fraction}) / \text{ADV}$	Pace estimate for early-session scans.
PM-to-PM RVOL	PM cumulative volume / average of stored prior PM totals	Closest to classic premarket RVOL, but chart/history dependent.
PM Score	$\text{abs}(\text{selected PM \%}) * \text{selected volume metric}$	Simple sort score to rank extreme gap plus volume participation.

5. Implementation Improvements

Upgrade	Why it matters
Pine Script v6	Aligns the script with the current Pine environment and future TradingView updates.
Screener-safe default volume mode	Avoids relying on a 20-day intraday PM array that Pine Screener may not have enough bars to build.
Direction-aware gap logic	Up movers can use PM high; down movers can use PM low; last-price mode remains available.
Weekday and day-reset hygiene	Prevents stale held values from carrying into a new session day before premarket starts.
Quality filters	Removes noisy symbols by price, PM volume, average daily volume, and PM dollar volume.

Separate up/down alerts	Lets you create long-side, short-side, or all-direction alerts without ambiguity.
Screener plot ordering	Optimizes the first ten columns Pine Screener initially adds to the table.
Dashboard and markers	Adds quick chart inspection without changing screener behavior.

Recommended default configuration

For premarket watchlist scans, start with the default PM volume / ADV mode, 4% gap threshold, 10% ADV threshold, latch enabled, and hold-after-open enabled. Then tune the liquidity filters to your universe. For a small-cap momentum list, a minimum PM dollar volume filter is usually more useful than price alone; for large caps, a minimum completed ADV filter can reduce noise.

6. Tutorial: Installation and Use

Step 1 - Install the script

- 1 Open TradingView and go to a chart.
- 2 Open Pine Editor.
- 3 Paste the contents of pm_movers_pro_screener_v2.pine.
- 4 Click Save, then Add to chart.
- 5 Open Chart Settings > Symbol and make sure extended-hours data is enabled for the symbol/data feed.

The script should be used on intraday timeframes. Pine Screener supports 1, 5, 15, 30, 60, 120, 240, 1D, 1W, and 1M intervals, but this particular script needs intraday bars to observe the premarket session.

Step 2 - Configure core inputs

Input	Default	Suggested use
Premarket session	0400-0930	Keep for Nasdaq/NYSE Arca style U.S. premarket. Change if your feed or strategy uses a different window.
Direction	Up	Use Both for a broader mover scan. Use Down for gap-down news scans.
Gap price basis	Extreme	Extreme is better for detecting a symbol that qualified at any point in PM. Last is stricter and requires the current/held PM price to qualify.
Volume filter mode	PM volume / ADV	Best first choice for Pine Screener. Switch to PM-to-PM RVOL only on charts with enough extended-hours history.
Latch signal	On	Recommended for scans. Once a symbol qualifies, it stays marked for that session day.
Hold after open	On	Recommended if scanning after 09:30 ET.

Step 3 - Use Pine Screener

- 1 Add the saved indicator to your TradingView favorites so Pine Screener can find it.
- 2 Open Products > Screeners > Pine.
- 3 Choose your watchlist.
- 4 Select PM Movers Pro as the indicator.
- 5 Choose an intraday timeframe. During premarket, 1m, 5m, or 15m are useful. After the open, 5m, 15m, or 30m often retain more relevant PM history in the last 500 bars.
- 6 Set filters: PM Mover (1/0) equals 1 and Config OK (1/0) equals 1.
- 7 Sort by PM Score, PM % Selected, PM Volume Metric, or PM Dollar Volume depending on your workflow.

Because Pine Screener initially adds the first ten plots as columns, the script publishes the most important scan fields first. Additional diagnostic fields, such as PM High %, PM Low %, PM Last %, PM Pace / ADV, and PM Avg Volume

Chart, are still available if you add them manually.

Step 4 - Create alerts

- 1 On a chart with the indicator applied, open the Alerts dialog.
- 2 Choose PM Movers Pro as the condition source.
- 3 For one-time session alerts, choose PM mover UP - first trigger or PM mover DOWN - first trigger.
- 4 For repeated qualifying bars, choose PM mover - every qualifying bar.
- 5 Use Once Per Bar Close if you want alerts to wait for each intraday bar to close.

Step 5 - Tune presets

Use case	Example settings	Notes
Large-cap news movers	Gap 2-4%; ADV % 0.03-0.10; min ADV 5M+	Large caps may have meaningful PM action at lower percent gaps.
Small-cap momentum	Gap 4-8%; ADV % 0.10-0.30; min PM dollar volume 250k-1M	Dollar volume is often more useful than share volume alone.
Gap-down watchlist	Direction Down; Extreme basis; gap 3-5%; min PM dollar volume	The upgraded script uses PM low for down-mover detection.
Very early PM scan	PM pace / ADV; minimum elapsed 5-15 minutes	Pace mode compensates for how little of the 04:00-09:30 window has elapsed.
Chart research	PM-to-PM RVOL; 15m/30m chart; extended history	Use this when you can visually confirm enough prior PM sessions are available.

7. Troubleshooting

Symptom	Likely cause	Fix
No signals during premarket	Extended-hours data is not enabled or not available on the data feed.	Enable extended hours on the chart. Confirm the symbol actually prints PM bars.
Config OK = 0	Non-intraday timeframe, missing daily data, or unavailable ADV.	Use 1m/5m/15m/30m/60m intraday and verify the symbol has daily volume history.
PM RVOL vs PM Avg is n/a	The chart-built PM average has not collected prior PM totals.	Use PM volume / ADV for Pine Screener or load a chart/timeframe with enough PM history.
Signal disappears after open	Hold after open is off or scan timeframe no longer contains the PM bars.	Turn Hold after open on. For later scans, use a higher intraday timeframe.

Too many low-quality names	Gap/volume thresholds are too permissive.	Raise ADV %, minimum PM dollar volume, minimum ADV, or minimum price.
Down movers do not appear	Direction is set to Up.	Set Direction to Down or Both.

8. Validation Notes and Limitations

Static validation was performed on the source file for structure, balanced delimiters, non-ASCII characters, and plot ordering. This environment does not include TradingView Pine Editor, so the final compile check should be performed by pasting the script into TradingView and using Pine Editor diagnostics.

Known limitations:

- No script can recover premarket bars that your plan, exchange subscription, or chart setting does not provide.
- ADV-based metrics are intentionally screener-safe, but they are not identical to true time-of-day relative volume.
- PM-to-PM RVOL is useful for chart research but depends on collecting prior PM session totals from available bars.
- Premarket liquidity can be thin. A screener signal is only a candidate-selection tool, not a trade decision.
- Pine Screener is watchlist-based. Results are only as broad as the symbols in the selected watchlist.

Risk note

FINRA highlights several extended-hours risks, including lower liquidity, greater volatility, unlinked markets, changing prices, news-announcement risk, and wider spreads. The script includes liquidity filters, but those filters do not remove market-structure risk.

9. Source References

- 1 TradingView Help Center, "TradingView Pine Screener: key features and requirements", <https://www.tradingview.com/support/solutions/43000742436-tradingview-pine-screener-key-features-and-requirements/>
- 2 TradingView Pine Script Docs, "Visuals / Overview", <https://www.tradingview.com/pine-script-docs/visuals/overview/>
- 3 TradingView Pine Script Docs, "Concepts / Sessions", <https://www.tradingview.com/pine-script-docs/concepts/sessions/>
- 4 TradingView Pine Script Docs, "Concepts / Time", <https://www.tradingview.com/pine-script-docs/concepts/time/>
- 5 TradingView Pine Script Docs, "Writing / Style guide", <https://www.tradingview.com/pine-script-docs/writing/style-guide/>
- 6 TradingView Pine Script Docs, "Language / Built-ins", <https://www.tradingview.com/pine-script-docs/language/built-ins/>
- 7 Nasdaq Listing Center, Nasdaq Equity 4 rules, <https://listingcenter.nasdaq.com/rulebook/nasdaq/rules/Nasdaq%20Equity%204>
- 8 NYSE, "Holidays & Trading Hours", <https://www.nyse.com/trade/hours-calendars>
- 9 FINRA, "Extended-Hours Trading: Know the Risks", <https://www.finra.org/investors/insights/extended-hours-trading>
- 10 FINRA Rule 2265, "Extended Hours Trading Risk Disclosure", <https://www.finra.org/rules-guidance/rulebooks/finra-rules/2265>

Appendix A: Full Pine Script v6 Source

Save the following as pm_movers_pro_screener_v2.pine or paste it directly into TradingView Pine Editor.

```
//@version=6
indicator("PM Movers Pro - Gap + Volume Screener", shorttitle = "PM Movers Pro", overlay =
true, max_labels_count = 100, precision = 2)

//-----
// PM Movers Pro
// Purpose: Find U.S. premarket gap movers with meaningful relative volume.
// Notes:
// - Built for intraday charts and TradingView Pine Screener.
// - Use extended-hours data. If your chart/data feed hides extended hours,
//   the script cannot see 04:00-09:30 ET bars.
// - The screener-safe volume modes use only two request.security() calls.
// - The PM-to-PM RVOL mode is chart/history dependent and may be unreliable
//   in Pine Screener on lower timeframes because the screener calculates a
//   limited number of bars.
//-----

//-----
// Inputs
//-----
const string GROUP_SESSION = "1. Session"
const string GROUP_SIGNAL = "2. Signal Logic"
const string GROUP_VOLUME = "3. Volume Filter"
const string GROUP_QUALITY = "4. Liquidity / Quality Filters"
const string GROUP_VISUALS = "5. Visuals"

string pmSessionInput = input.session("0400-0930", "Premarket session", group = GROUP_SESSION,
tooltip = "Default is the common U.S. extended-hours premarket window. Adjust if your venue,
broker, or data feed uses a different window.")
string tzInput = input.string("America/New_York", "Session timezone", group = GROUP_
SESSION, tooltip = "Use an IANA timezone string. America/New_York automatically handles
daylight saving time.")
bool weekdaysOnlyInput = input.bool(true, "Restrict to Monday-Friday", group = GROUP_SESSION)
bool holdAfterOpenInput = input.bool(true, "Keep today's PM values after 09:30", group =
GROUP_SESSION, tooltip = "Useful for watchlists and Pine Screener after the opening bell.
Values reset on the next session day.")
int pmFullMinutesInput = input.int(330, "Full PM session length, minutes", minval = 1, maxval =
1440, group = GROUP_SESSION, tooltip = "Used only by the pace-based volume mode. Default
330 minutes = 04:00-09:30 ET.")
int minElapsedMinutesInput = input.int(0, "Minimum elapsed PM minutes before signal", minval =
0, maxval = 1440, group = GROUP_SESSION)

float gapThresholdPctInput = input.float(4.0, "Gap threshold, %", minval = 0.0, step = 0.1,
group = GROUP_SIGNAL)
string directionInput = input.string("Up", "Direction", options = ["Up", "Down", "Both"],
group = GROUP_SIGNAL)
string priceBasisInput = input.string("Extreme (direction-aware)", "Gap price basis", options
= ["Extreme (direction-aware)", "Last"], group = GROUP_SIGNAL, tooltip = "Extreme uses PM
high for up movers and PM low for down movers. Last uses current/last PM price.")
bool latchSignalInput = input.bool(true, "Latch once triggered for the day", group = GROUP_
SIGNAL, tooltip = "Once true, the signal remains true until the next session day. This is
usually best for screener scans.")

int lookbackDaysInput = input.int(20, "Completed daily volume lookback", minval = 2, maxval =
200, group = GROUP_VOLUME)
string volumeModeInput = input.string("PM volume / ADV (screener-safe)", "Volume filter mode",
options = ["PM volume / ADV (screener-safe)", "PM pace / ADV (screener-safe)", "PM-to-PM
RVOL (chart/history dependent)"], group = GROUP_VOLUME)
float advPctThresholdInput = input.float(0.10, "ADV % threshold for screener-safe modes",
minval = 0.0, step = 0.01, group = GROUP_VOLUME, tooltip = "0.10 means today's premarket
volume must be at least 10% of the completed daily average volume.")
float pmRvolThresholdInput = input.float(2.0, "PM-to-PM RVOL threshold", minval = 0.0, step =
0.1, group = GROUP_VOLUME, tooltip = "Used only when Volume filter mode is PM-to-PM RVOL.")

float minPriceInput = input.float(0.0, "Minimum PM last price", minval = 0.0, step = 0.01,
group = GROUP_QUALITY, tooltip = "0 disables this filter.")
float minPmVolumeInput = input.float(0.0, "Minimum PM cumulative volume", minval = 0.0, step =
1000.0, group = GROUP_QUALITY, tooltip = "0 disables this filter.")
float minAvgDailyVolumeInput = input.float(0.0, "Minimum completed ADV", minval = 0.0, step =
10000.0, group = GROUP_QUALITY, tooltip = "0 disables this filter.")
float minPmDollarVolInput = input.float(0.0, "Minimum PM dollar volume", minval = 0.0, step =
10000.0, group = GROUP_QUALITY, tooltip = "Uses PM cumulative volume multiplied by the PM
last price. 0 disables this filter.")

bool showMarkersInput = input.bool(true, "Show chart markers", group = GROUP_VISUALS)
bool highlightBgInput = input.bool(false, "Highlight signal background", group = GROUP_
VISUALS)
bool showDashboardInput = input.bool(true, "Show last-bar dashboard", group = GROUP_VISUALS)

//-----
// Higher-timeframe reference data. These two calls keep the script under Pine
// Screener's request.*() limit and use supported 1D data.
//-----
float prevClose = request.security(syminfo.tickerid, "1D", close[1], gaps = barmerge.gaps_off,
lookahead = barmerge.lookahead_on, ignore_invalid_symbol = true)
float avgDailyVol = request.security(syminfo.tickerid, "1D", ta.sma(volume, lookbackDaysInput)
[1], gaps = barmerge.gaps_off, lookahead = barmerge.lookahead_on, ignore_invalid_symbol =
```

```

true)

//-----
// Session detection
//-----
bool inWeekday = not na(time(timeframe.period, "0000-0000:23456", tzInput))
bool inPmRaw = not na(time(timeframe.period, pmSessionInput, tzInput))
bool inPm = inPmRaw and (not weekdaysOnlyInput or inWeekday)
bool pmStart = inPm and not inPm[1]
bool pmEnd = not inPm and inPm[1]

int sessionDayKey = year(time, tzInput) * 10000 + month(time, tzInput) * 100 + dayofmonth(
time, tzInput)
bool newSessionDay = na(sessionDayKey[1]) or sessionDayKey != sessionDayKey[1]

//-----
// Track today's premarket stats
//-----
var float pmCumVol := na
var float pmHigh := na
var float pmLow := na
var float pmLast := na
var int pmStartTime := na
var bool latchedUp := false
var bool latchedDown := false

if newSessionDay
    pmCumVol := na
    pmHigh := na
    pmLow := na
    pmLast := na
    pmStartTime := na
    latchedUp := false
    latchedDown := false

float barVolume := nz(volume, 0.0)

if pmStart
    pmCumVol := barVolume
    pmHigh := high
    pmLow := low
    pmLast := close
    pmStartTime := time
else if inPm
    pmCumVol := na(pmCumVol) ? barVolume : pmCumVol + barVolume
    pmHigh := na(pmHigh) ? high : math.max(pmHigh, high)
    pmLow := na(pmLow) ? low : math.min(pmLow, low)
    pmLast := close

float pmVolForDay = inPm ? pmCumVol : (holdAfterOpenInput ? pmCumVol : na)
float pmHighForDay = inPm ? pmHigh : (holdAfterOpenInput ? pmHigh : na)
float pmLowForDay = inPm ? pmLow : (holdAfterOpenInput ? pmLow : na)
float pmLastForDay = inPm ? pmLast : (holdAfterOpenInput ? pmLast : na)

int barCloseTime = na(time_close) ? time : time_close
float elapsedMinutesActive = inPm and not na(pmStartTime) ? math.max(1.0, (barCloseTime -
pmStartTime) / 60000.0) : na
float elapsedMinutesForDay = inPm ? elapsedMinutesActive : (holdAfterOpenInput and not na(
pmVolForDay) ? pmFullMinutesInput : na)
float elapsedFraction = na(elapsedMinutesForDay) ? na : math.min(1.0, math.max(
elapsedMinutesForDay / pmFullMinutesInput, 0.01))

//-----
// Chart/history dependent premarket average. Useful on charts with enough
// extended-hours bars; not the default screener-safe baseline.
//-----
var array<float> pmTotals = array.new_float()

if pmEnd and not na(pmCumVol) and pmCumVol > 0
    array.unshift(pmTotals, pmCumVol)
    if array.size(pmTotals) > lookbackDaysInput
        array.pop(pmTotals)

float pmAvgVolChart = array.size(pmTotals) > 0 ? array.sum(pmTotals) / array.size(pmTotals) :
na

//-----
// Price, volume, and quality metrics
//-----
float pmHighPct = na(prevClose) or na(pmHighForDay) or prevClose == 0.0 ? na : (pmHighForDay -
prevClose) / prevClose * 100.0
float pmLowPct = na(prevClose) or na(pmLowForDay) or prevClose == 0.0 ? na : (pmLowForDay -
prevClose) / prevClose * 100.0
float pmLastPct = na(prevClose) or na(pmLastForDay) or prevClose == 0.0 ? na : (pmLastForDay -
prevClose) / prevClose * 100.0

float selectedPct = directionInput == "Up" ? (priceBasisInput == "Last" ? pmLastPct :
pmHighPct) :
    directionInput == "Down" ? (priceBasisInput == "Last" ? pmLastPct : pmLowPct) :
    priceBasisInput == "Last" ? pmLastPct :
    math.abs(nz(pmHighPct, 0.0)) >= math.abs(nz(pmLowPct, 0.0)) ? pmHighPct : pmLowPct

float pmVolToAdv = na(pmVolForDay) or na(avgDailyVol) or avgDailyVol <= 0.0 ? na :
pmVolForDay / avgDailyVol

```

```

float pmPaceToAdv = na(pmVolToAdv) or na(elapsedFraction) or elapsedFraction <= 0.0 ? na :
pmVolToAdv / elapsedFraction
float pmRvol = na(pmVolForDay) or na(pmAvgVolChart) or pmAvgVolChart <= 0.0 ? na :
pmVolForDay / pmAvgVolChart
float pmDollarVolume = na(pmVolForDay) or na(pmLastForDay) ? na : pmVolForDay * pmLastForDay

float volumeMetric = switch volumeModeInput
  "PM-to-PM RVOL (chart/history dependent)" => pmRvol
  "PM pace / ADV (screener-safe)" => pmPaceToAdv
  => pmVolToAdv

bool volumeOk = switch volumeModeInput
  "PM-to-PM RVOL (chart/history dependent)" => not na(pmRvol) and pmRvol >=
pmRvolThresholdInput
  => not na(volumeMetric) and volumeMetric >= advPctThresholdInput

bool directionAllowsUp = directionInput == "Up" or directionInput == "Both"
bool directionAllowsDown = directionInput == "Down" or directionInput == "Both"
bool gapUpOk = priceBasisInput == "Last" ? not na(pmLastPct) and pmLastPct >=
gapThresholdPctInput : not na(pmHighPct) and pmHighPct >= gapThresholdPctInput
bool gapDownOk = priceBasisInput == "Last" ? not na(pmLastPct) and pmLastPct <= -
gapThresholdPctInput : not na(pmLowPct) and pmLowPct <= -gapThresholdPctInput

bool timeframeOk = timeframe.isintraday
bool elapsedOk = minElapsedMinutesInput == 0 or not na(elapsedMinutesForDay) and
elapsedMinutesForDay >= minElapsedMinutesInput
bool priceOk = minPriceInput <= 0.0 or not na(pmLastForDay) and pmLastForDay >= minPriceInput
bool pmVolumeOk = minPmVolumeInput <= 0.0 or not na(pmVolForDay) and pmVolForDay >=
minPmVolumeInput
bool avgDailyVolumeOk = minAvgDailyVolumeInput <= 0.0 or not na(avgDailyVol) and avgDailyVol
>= minAvgDailyVolumeInput
bool dollarVolumeOk = minPmDollarVolInput <= 0.0 or not na(pmDollarVolume) and pmDollarVolume
>= minPmDollarVolInput
bool configOk = timeframeOk and not na(prevClose) and not na(avgDailyVol)
bool qualityOk = configOk and elapsedOk and priceOk and pmVolumeOk and avgDailyVolumeOk and
dollarVolumeOk

bool moverUpNow = inPm and qualityOk and volumeOk and directionAllowsUp and gapUpOk
bool moverDownNow = inPm and qualityOk and volumeOk and directionAllowsDown and gapDownOk
bool moverNow = moverUpNow or moverDownNow

//-----
// Latching logic
//-----
bool moverUpFirst = false
bool moverDownFirst = false

if latchSignalInput
  moverUpFirst := moverUpNow and not latchedUp
  moverDownFirst := moverDownNow and not latchedDown
  if moverUpNow
    latchedUp := true
  if moverDownNow
    latchedDown := true
else
  moverUpFirst := moverUpNow and not moverUpNow[1]
  moverDownFirst := moverDownNow and not moverDownNow[1]

bool upForScreener = latchSignalInput ? (holdAfterOpenInput ? latchedUp : inPm and latchedUp)
:
  (holdAfterOpenInput ? qualityOk and volumeOk and directionAllowsUp and gapUpOk :
moverUpNow)
bool downForScreener = latchSignalInput ? (holdAfterOpenInput ? latchedDown : inPm and
latchedDown) :
  (holdAfterOpenInput ? qualityOk and volumeOk and directionAllowsDown and gapDownOk :
moverDownNow)
bool moverForScreener = upForScreener or downForScreener

int signalDirection = upForScreener and not downForScreener ? 1 :
  downForScreener and not upForScreener ? -1 :
  selectedPct >= 0 ? 1 : -1
float signalScore = na(selectedPct) or na(volumeMetric) ? na : math.abs(selectedPct) *
volumeMetric

//-----
// Screener and Data Window outputs. Pine Screener initially adds the first ten
// plots as columns, so the most important columns are listed first.
//-----
plot(moverForScreener ? 1 : 0, title = "PM Mover (1/0)", display = display.data_window)
plot(signalDirection, title = "PM Direction (+1/-1)", display = display.data_window)
plot(selectedPct, title = "PM % Selected", display = display.data_window)
plot(volumeMetric, title = "PM Volume Metric", display = display.data_window)
plot(pmVolToAdv, title = "PM Vol / ADV", display = display.data_window)
plot(pmRvol, title = "PM RVOL vs PM Avg", display = display.data_window)
plot(pmVolForDay, title = "PM Cum Volume", display = display.data_window)
plot(pmDollarVolume, title = "PM Dollar Volume", display = display.data_window)
plot(signalScore, title = "PM Score", display = display.data_window)
plot(configOk ? 1 : 0, title = "Config OK (1/0)", display = display.data_window)
plot(pmLastPct, title = "PM Last %", display = display.data_window)
plot(pmHighPct, title = "PM High %", display = display.data_window)
plot(pmLowPct, title = "PM Low %", display = display.data_window)
plot(pmPaceToAdv, title = "PM Pace / ADV", display = display.data_window)
plot(pmAvgVolChart, title = "PM Avg Volume Chart", display = display.data_window)
//-----

```

```
// Visuals
//-----
plotshape(showMarkersInput and moverUpNow, title = "PM mover up", style = shape.triangleup,
location = location.belowbar, size = size.tiny, text = "PM+", color = color.new(color.lime, 0)
, textcolor = color.white)
plotshape(showMarkersInput and moverDownNow, title = "PM mover down", style = shape.
triangledown, location = location.abovebar, size = size.tiny, text = "PM-", color = color.new(
color.red, 0), textcolor = color.white)
bgcolor(highlightBgInput and moverUpNow ? color.new(color.green, 88) : highlightBgInput and
moverDownNow ? color.new(color.red, 88) : na)

fmtPct(float x) =>
na(x) ? "n/a" : str.toString(x, "%.##") + "%"

fmtRatio(float x) =>
na(x) ? "n/a" : str.toString(x, "%.###")

fmtVol(float x) =>
na(x) ? "n/a" : str.toString(x, format.volume)

var table dashTable = table.new(position.top_right, 2, 8, border_width = 1)

if showDashboardInput and barstate.islast
string directionText = moverForScreener ? (signalDirection == 1 ? "UP" : "DOWN") : "No
signal"
color directionBg = moverForScreener ? (signalDirection == 1 ? color.new(color.green, 80)
: color.new(color.red, 80)) : color.new(color.gray, 85)
table.cell(dashTable, 0, 0, "PM Movers Pro", text_color = chart.fg_color, bgcolor = color.
new(color.blue, 85))
table.cell(dashTable, 1, 0, directionText, text_color = chart.fg_color, bgcolor =
directionBg)
table.cell(dashTable, 0, 1, "PM %", text_color = chart.fg_color)
table.cell(dashTable, 1, 1, fmtPct(selectedPct), text_color = chart.fg_color)
table.cell(dashTable, 0, 2, "Vol metric", text_color = chart.fg_color)
table.cell(dashTable, 1, 2, fmtRatio(volumeMetric), text_color = chart.fg_color)
table.cell(dashTable, 0, 3, "PM Vol", text_color = chart.fg_color)
table.cell(dashTable, 1, 3, fmtVol(pmVolForDay), text_color = chart.fg_color)
table.cell(dashTable, 0, 4, "PM $ Vol", text_color = chart.fg_color)
table.cell(dashTable, 1, 4, fmtVol(pmDollarVolume), text_color = chart.fg_color)
table.cell(dashTable, 0, 5, "ADV", text_color = chart.fg_color)
table.cell(dashTable, 1, 5, fmtVol(avgDailyVol), text_color = chart.fg_color)
table.cell(dashTable, 0, 6, "Elapsed", text_color = chart.fg_color)
table.cell(dashTable, 1, 6, na(elapsedMinutesForDay) ? "n/a" : str.toString(
elapsedMinutesForDay, "%.0") + "m", text_color = chart.fg_color)
table.cell(dashTable, 0, 7, "Config", text_color = chart.fg_color)
table.cell(dashTable, 1, 7, configOk ? "OK" : "Check TF/data", text_color = chart.fg_
color)

//-----
// Alerts
//-----
alertcondition(moverUpFirst, title = "PM mover UP - first trigger", message = "PM mover UP
first trigger: {{ticker}} | PM%={{plot(\"PM % Selected\")}} | VolMetric={{plot(\"PM Volume
Metric\")}} | PMVol={{plot(\"PM Cum Volume\")}}")
alertcondition(moverDownFirst, title = "PM mover DOWN - first trigger", message = "PM mover
DOWN first trigger: {{ticker}} | PM%={{plot(\"PM % Selected\")}} | VolMetric={{plot(\"PM
Volume Metric\")}} | PMVol={{plot(\"PM Cum Volume\")}}")
alertcondition(moverNow, title = "PM mover - every qualifying bar", message = "PM mover
qualifying bar: {{ticker}} | Dir={{plot(\"PM Direction (+1/-1)\")}} | PM%={{plot(\"PM %
Selected\")}} | VolMetric={{plot(\"PM Volume Metric\")}} | PMVol={{plot(\"PM Cum Volume\")}}")
```